

Definite integrals – 5

Integral identities and what they force upon the function

Homework

11 сентября 2026 года

Theoretical minimum. Moments determine a continuous function, because polynomials are uniformly dense; finitely many vanishing moments force sign changes. An identity holding for all x is to be differentiated, and a window $[x-1, x]$ or $[0, \lambda x]$ turns it into a functional equation for f ; two windows of incommensurable lengths leave almost no freedom. An oscillating factor averages out: $\int f(x)g(nx)dx \rightarrow \int f \cdot \int g$ for 1-periodic g , and by Weyl's criterion it is enough to test equidistribution on powers x^k . Monotone or concave weights are compared by Chebyshev's integral inequality; ratios of the form $\int f^{n+1}/g^n$ are handled by Jensen or by Hölder with the exponents read off from the powers (Inequalities – 4). Finally, an oscillatory integral is bounded by the second mean value theorem, and the true order of decay is found by integrating by parts twice (Definite integrals – 6).

1. (Putnam 2017) Let a and b be real numbers with $a < b$, and let f and g be continuous functions from $[a, b]$ to $(0, \infty)$ such that $\int_a^b f(x)dx = \int_a^b g(x)dx$ but $f \neq g$. For every positive integer n , define

$$I_n = \int_a^b \frac{(f(x))^{n+1}}{(g(x))^n} dx.$$

Show that $I_1, I_2, I_3, \dots$ is an increasing sequence with $\lim_{n \rightarrow \infty} I_n = \infty$.

2. (VJIMC 2001) Let $f: [0, 1] \rightarrow \mathbb{R}$ be a continuous function. Define a sequence of functions $f_n: [0, 1] \rightarrow \mathbb{R}$ in the following way:

$$f_0(x) = f(x), \quad f_{n+1}(x) = \int_0^x f_n(t)dt, \quad n = 0, 1, 2, \dots$$

Prove that if $f_n(1) = 0$ for all n , then $f(x) = 0$.

3. (VJIMC 2003) Let $f, g: [0, 1] \rightarrow (0, +\infty)$ be two continuous functions such that f and $\frac{g}{f}$ are increasing. Prove that

$$\int_0^1 \frac{\int_0^x f(t)dt}{\int_0^x g(t)dt} dx \leq 2 \int_0^1 \frac{f(t)}{g(t)} dt.$$

4. (VJIMC 2010) Prove that there exist positive constants c_1 and c_2 with the following properties:

- a) For all real $k > 1$, $\left| \int_0^1 \sqrt{1-x^2} \cos(kx) dx \right| < \frac{c_1}{k^{3/2}}.$
b) For all real $k > 1$, $\left| \int_0^1 \sqrt{1-x^2} \sin(kx) dx \right| > \frac{c_2}{k}.$

5. (VJIMC 2016) Let $f: [0, \infty) \rightarrow \mathbb{R}$ be a continuously differentiable function satisfying

$$f(x) = \int_{x-1}^x f(t)dt$$

for all $x \geq 1$. Show that f has bounded variation on $[1, \infty)$, i.e. $\int_1^\infty |f'(x)|dx < \infty$.

6. (SEEMOUS 2017) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Prove that

$$\int_0^4 f(x(x-3)^2)dx = 2 \int_1^3 f(x(x-3)^2)dx.$$

7. (SEEMOUS 2019) A sequence $(x_n)_{n \geq 1}$ of numbers in the interval $[0, 1]$ is called a Devin sequence if for every continuous function $f: [0, 1] \rightarrow \mathbb{R}$ the following relation holds:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n f(x_i) = \int_0^1 f(x)dx.$$

Show that a sequence $(x_n)_{n \geq 1}$ of numbers in $[0, 1]$ is a Devin sequence if and only if $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n x_i^k = \frac{1}{k+1}$ for every k .

8. (RMO 2011) Let $f: [0, 1] \rightarrow (0, \infty)$ be a continuous function having the property that for any natural number $n \geq 2$ there exist $n-1$ real

numbers $0 < t_1 < t_2 < \cdots < t_{n-1} < 1$ such that

$$\int_0^{t_1} f(t)dt = \int_{t_1}^{t_2} f(t)dt = \cdots = \int_{t_{n-2}}^{t_{n-1}} f(t)dt = \int_{t_{n-1}}^1 f(t)dt.$$

Calculate

$$\lim_{n \rightarrow \infty} \frac{n}{\frac{1}{f(0)} + \sum_{i=1}^{n-1} \frac{1}{f(t_i)} + \frac{1}{f(1)}}.$$

9. (RMO 1999) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a monotonic function and a, b, c, d be real numbers with a and c nonzero. Prove that if the equalities

$$\int_x^{x+\sqrt{3}} f(t)dt = ax + b \quad \text{and} \quad \int_x^{x+\sqrt{2}} f(t)dt = cx + d$$

hold for every real number x , then f is a polynomial function of degree one.

10. (RMO 1998) Suppose that $a, b \in \mathbb{R}_+$ with $a+b < 1$ and let $f: [0, +\infty) \rightarrow [0, +\infty)$ be an increasing function such that

$$\int_0^x f(t)dt = \int_0^{ax} f(t)dt + \int_0^{bx} f(t)dt \quad \text{for all } x \geq 0.$$

Prove that $f(x) = 0$ for all $x \geq 0$.